# Pandas toolkit Part 5

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In [1]:
```
import pandas as pd
import numpy as np
```

In [4]:
```
url = ("https://raw.github.com/pandas-dev""/pandas/main/pandas/tests/io/data/cs
tips = pd.read_csv(url)
tips.head()
```

Out[4]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| **0** | 16.99 | 1.01 | Female | No | Sun | Dinner | 2 |
| **1** | 10.34 | 1.66 | Male | No | Sun | Dinner | 3 |
| **2** | 21.01 | 3.50 | Male | No | Sun | Dinner | 3 |
| **3** | 23.68 | 3.31 | Male | No | Sun | Dinner | 2 |
| **4** | 24.59 | 3.61 | Female | No | Sun | Dinner | 4 |

In [3]:
```
tips[["total_bill", "tip", "smoker", "time"]]
```

Out[3]:

| | total_bill | tip | smoker | time |
|---|---|---|---|---|
| **0** | 16.99 | 1.01 | No | Dinner |
| **1** | 10.34 | 1.66 | No | Dinner |
| **2** | 21.01 | 3.50 | No | Dinner |
| **3** | 23.68 | 3.31 | No | Dinner |
| **4** | 24.59 | 3.61 | No | Dinner |
| **...** | ... | ... | ... | ... |
| **239** | 29.03 | 5.92 | No | Dinner |
| **240** | 27.18 | 2.00 | Yes | Dinner |
| **241** | 22.67 | 2.00 | Yes | Dinner |
| **242** | 17.82 | 1.75 | No | Dinner |
| **243** | 18.78 | 3.00 | No | Dinner |

244 rows × 4 columns

In [4]:
```
tips.assign(tip_rate=tips["tip"] / tips["total_bill"])
```

Out[4]:

| | total_bill | tip | sex | smoker | day | time | size | tip_rate |
|---|---|---|---|---|---|---|---|---|
| **0** | 16.99 | 1.01 | Female | No | Sun | Dinner | 2 | 0.059447 |
| **1** | 10.34 | 1.66 | Male | No | Sun | Dinner | 3 | 0.160542 |
| **2** | 21.01 | 3.50 | Male | No | Sun | Dinner | 3 | 0.166587 |
| **3** | 23.68 | 3.31 | Male | No | Sun | Dinner | 2 | 0.139780 |
| **4** | 24.59 | 3.61 | Female | No | Sun | Dinner | 4 | 0.146808 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... |
| **239** | 29.03 | 5.92 | Male | No | Sat | Dinner | 3 | 0.203927 |
| **240** | 27.18 | 2.00 | Female | Yes | Sat | Dinner | 2 | 0.073584 |
| **241** | 22.67 | 2.00 | Male | Yes | Sat | Dinner | 2 | 0.088222 |
| **242** | 17.82 | 1.75 | Male | No | Sat | Dinner | 2 | 0.098204 |
| **243** | 18.78 | 3.00 | Female | No | Thur | Dinner | 2 | 0.159744 |

244 rows × 8 columns

In [5]:
```
is_dinner = tips["time"] == "Dinner"
is_dinner
```

Out[5]:
```
0      True
1      True
2      True
3      True
4      True
       ...
239    True
240    True
241    True
242    True
243    True
Name: time, Length: 244, dtype: bool
```

In [6]:
```
is_dinner.value_counts()
```

Out[6]:
```
True     176
False     68
Name: time, dtype: int64
```

In [7]:
```
tips[is_dinner]
```

Out[7]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| **0** | 16.99 | 1.01 | Female | No | Sun | Dinner | 2 |
| **1** | 10.34 | 1.66 | Male | No | Sun | Dinner | 3 |
| **2** | 21.01 | 3.50 | Male | No | Sun | Dinner | 3 |
| **3** | 23.68 | 3.31 | Male | No | Sun | Dinner | 2 |
| **4** | 24.59 | 3.61 | Female | No | Sun | Dinner | 4 |
| **...** | ... | ... | ... | ... | ... | ... | ... |
| **239** | 29.03 | 5.92 | Male | No | Sat | Dinner | 3 |
| **240** | 27.18 | 2.00 | Female | Yes | Sat | Dinner | 2 |
| **241** | 22.67 | 2.00 | Male | Yes | Sat | Dinner | 2 |
| **242** | 17.82 | 1.75 | Male | No | Sat | Dinner | 2 |
| **243** | 18.78 | 3.00 | Female | No | Thur | Dinner | 2 |

176 rows × 7 columns

In [8]:
```
tips[(tips["time"] == "Dinner") & (tips["tip"] > 5.00)]
```

Out[8]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| **23** | 39.42 | 7.58 | Male | No | Sat | Dinner | 4 |
| **44** | 30.40 | 5.60 | Male | No | Sun | Dinner | 4 |
| **47** | 32.40 | 6.00 | Male | No | Sun | Dinner | 4 |
| **52** | 34.81 | 5.20 | Female | No | Sun | Dinner | 4 |
| **59** | 48.27 | 6.73 | Male | No | Sat | Dinner | 4 |
| **116** | 29.93 | 5.07 | Male | No | Sun | Dinner | 4 |
| **155** | 29.85 | 5.14 | Female | No | Sun | Dinner | 5 |
| **170** | 50.81 | 10.00 | Male | Yes | Sat | Dinner | 3 |
| **172** | 7.25 | 5.15 | Male | Yes | Sun | Dinner | 2 |
| **181** | 23.33 | 5.65 | Male | Yes | Sun | Dinner | 2 |
| **183** | 23.17 | 6.50 | Male | Yes | Sun | Dinner | 4 |
| **211** | 25.89 | 5.16 | Male | Yes | Sat | Dinner | 4 |
| **212** | 48.33 | 9.00 | Male | No | Sat | Dinner | 4 |
| **214** | 28.17 | 6.50 | Female | Yes | Sat | Dinner | 3 |
| **239** | 29.03 | 5.92 | Male | No | Sat | Dinner | 3 |

In [9]:
```
tips[(tips["size"] >= 5) | (tips["total_bill"] > 45)]
```

Out[9]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| **59** | 48.27 | 6.73 | Male | No | Sat | Dinner | 4 |
| **125** | 29.80 | 4.20 | Female | No | Thur | Lunch | 6 |
| **141** | 34.30 | 6.70 | Male | No | Thur | Lunch | 6 |
| **142** | 41.19 | 5.00 | Male | No | Thur | Lunch | 5 |
| **143** | 27.05 | 5.00 | Female | No | Thur | Lunch | 6 |
| **155** | 29.85 | 5.14 | Female | No | Sun | Dinner | 5 |
| **156** | 48.17 | 5.00 | Male | No | Sun | Dinner | 6 |
| **170** | 50.81 | 10.00 | Male | Yes | Sat | Dinner | 3 |
| **182** | 45.35 | 3.50 | Male | Yes | Sun | Dinner | 3 |
| **185** | 20.69 | 5.00 | Male | No | Sun | Dinner | 5 |
| **187** | 30.46 | 2.00 | Male | Yes | Sun | Dinner | 5 |
| **212** | 48.33 | 9.00 | Male | No | Sat | Dinner | 4 |
| **216** | 28.15 | 3.00 | Male | Yes | Sat | Dinner | 5 |

In [10]:
```
tips.groupby("sex").size()
```

Out[10]:
```
sex
Female     87
Male      157
dtype: int64
```

In [11]:
```
tips.groupby("sex").count()
```

Out[11]:

| | total_bill | tip | smoker | day | time | size |
|---|---|---|---|---|---|---|
| **sex** | | | | | | |
| **Female** | 87 | 87 | 87 | 87 | 87 | 87 |
| **Male** | 157 | 157 | 157 | 157 | 157 | 157 |

In [12]:
```
tips.groupby("sex")["total_bill"].count()
```

Out[12]:
```
sex
Female     87
Male      157
Name: total_bill, dtype: int64
```

In [13]:
```
tips.groupby("day").agg({"tip": np.mean, "day": np.size})
```

Out[13]:

| | tip | day |
|---|---|---|
| **day** | | |
| **Fri** | 2.734737 | 19 |
| **Sat** | 2.993103 | 87 |
| **Sun** | 3.255132 | 76 |
| **Thur** | 2.771452 | 62 |

In [14]:
```
tips.groupby(["smoker", "day"]).agg({"tip": [np.size, np.mean]})
```

Out[14]:

| | | tip | |
|---|---|---|---|
| | | **size** | **mean** |
| **smoker** | **day** | | |
| **No** | **Fri** | 4 | 2.812500 |
| | **Sat** | 45 | 3.102889 |
| | **Sun** | 57 | 3.167895 |
| | **Thur** | 45 | 2.673778 |
| **Yes** | **Fri** | 15 | 2.714000 |
| | **Sat** | 42 | 2.875476 |
| | **Sun** | 19 | 3.516842 |
| | **Thur** | 17 | 3.030000 |

In [15]:
```
tips.nlargest(10 + 5, columns="tip").tail(2)
```

Out[15]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| **85** | 34.83 | 5.17 | Female | No | Thur | Lunch | 4 |
| **211** | 25.89 | 5.16 | Male | Yes | Sat | Dinner | 4 |

In [17]:
```python
(
tips.assign(
rn=tips.sort_values(["total_bill"], ascending=False)
.groupby(["day"])
.cumcount()
+ 1
)
.query("rn < 3")
.sort_values(["day", "rn"])
)
```

Out[17]:

| | total_bill | tip | sex | smoker | day | time | size | rn |
|---|---|---|---|---|---|---|---|---|
| **95** | 40.17 | 4.73 | Male | Yes | Fri | Dinner | 4 | 1 |
| **90** | 28.97 | 3.00 | Male | Yes | Fri | Dinner | 2 | 2 |
| **170** | 50.81 | 10.00 | Male | Yes | Sat | Dinner | 3 | 1 |
| **212** | 48.33 | 9.00 | Male | No | Sat | Dinner | 4 | 2 |
| **156** | 48.17 | 5.00 | Male | No | Sun | Dinner | 6 | 1 |
| **182** | 45.35 | 3.50 | Male | Yes | Sun | Dinner | 3 | 2 |
| **197** | 43.11 | 5.00 | Female | Yes | Thur | Lunch | 4 | 1 |
| **142** | 41.19 | 5.00 | Male | No | Thur | Lunch | 5 | 2 |

In [18]:
```python
(
tips.assign(
rnk=tips.groupby(["day"])["total_bill"].rank(
method="first", ascending=False
)
)
.query("rnk < 3")
.sort_values(["day", "rnk"])
)
```

Out[18]:

| | total_bill | tip | sex | smoker | day | time | size | rnk |
|---|---|---|---|---|---|---|---|---|
| **95** | 40.17 | 4.73 | Male | Yes | Fri | Dinner | 4 | 1.0 |
| **90** | 28.97 | 3.00 | Male | Yes | Fri | Dinner | 2 | 2.0 |
| **170** | 50.81 | 10.00 | Male | Yes | Sat | Dinner | 3 | 1.0 |
| **212** | 48.33 | 9.00 | Male | No | Sat | Dinner | 4 | 2.0 |
| **156** | 48.17 | 5.00 | Male | No | Sun | Dinner | 6 | 1.0 |
| **182** | 45.35 | 3.50 | Male | Yes | Sun | Dinner | 3 | 2.0 |
| **197** | 43.11 | 5.00 | Female | Yes | Thur | Lunch | 4 | 1.0 |
| **142** | 41.19 | 5.00 | Male | No | Thur | Lunch | 5 | 2.0 |

In [19]:
```python
(
tips[tips["tip"] < 2]
.assign(rnk_min=tips.groupby(["sex"])["tip"].rank(method="min"))
.query("rnk_min < 3")
.sort_values(["sex", "rnk_min"])
)
```

Out[19]:

| | total_bill | tip | sex | smoker | day | time | size | rnk_min |
|---|---|---|---|---|---|---|---|---|
| **67** | 3.07 | 1.00 | Female | Yes | Sat | Dinner | 1 | 1.0 |
| **92** | 5.75 | 1.00 | Female | Yes | Fri | Dinner | 2 | 1.0 |
| **111** | 7.25 | 1.00 | Female | No | Sat | Dinner | 1 | 1.0 |
| **236** | 12.60 | 1.00 | Male | Yes | Sat | Dinner | 2 | 1.0 |
| **237** | 32.83 | 1.17 | Male | Yes | Sat | Dinner | 2 | 2.0 |

In [20]:
```python
tips.loc[tips["tip"] < 2, "tip"] *= 2
```

In [21]:
```python
tips = tips.loc[tips["tip"] <= 9]
```

In [23]:
```python
tips = pd.read_csv("tips.csv", sep="\t", header=None)
# alternatively, read_table is an alias to read_csv with tab delimiter
tips = pd.read_table("tips.csv", header=None)
```

In [24]:
```python
tips.to_excel("./tips.xlsx")
```

In [25]:
```python
tips_df = pd.read_excel("./tips.xlsx", index_col=0)
```

In [26]:
```python
tips.head(5)
```

Out[26]:

| | 0 |
|---|---|
| **0** | total_bill,tip,sex,smoker,day,time,size |
| **1** | 16.99,1.01,Female,No,Sun,Dinner,2 |
| **2** | 10.34,1.66,Male,No,Sun,Dinner,3 |
| **3** | 21.01,3.5,Male,No,Sun,Dinner,3 |
| **4** | 23.68,3.31,Male,No,Sun,Dinner,2 |

In [27]:
```python
tips = pd.read_csv("tips.csv", sep="\t", header=None)
# alternatively, read_table is an alias to read_csv with tab delimiter
tips = pd.read_table("tips.csv", header=None)
tips.head()
```

Out[27]:

| | 0 |
|---|---|
| 0 | total_bill,tip,sex,smoker,day,time,size |
| 1 | 16.99,1.01,Female,No,Sun,Dinner,2 |
| 2 | 10.34,1.66,Male,No,Sun,Dinner,3 |
| 3 | 21.01,3.5,Male,No,Sun,Dinner,3 |
| 4 | 23.68,3.31,Male,No,Sun,Dinner,2 |

In [28]:
```python
url = ("https://raw.github.com/pandas-dev""/pandas/main/pandas/tests/io/data/cs

tips = pd.read_csv(url)
tips.head()
```

Out[28]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| 0 | 16.99 | 1.01 | Female | No | Sun | Dinner | 2 |
| 1 | 10.34 | 1.66 | Male | No | Sun | Dinner | 3 |
| 2 | 21.01 | 3.50 | Male | No | Sun | Dinner | 3 |
| 3 | 23.68 | 3.31 | Male | No | Sun | Dinner | 2 |
| 4 | 24.59 | 3.61 | Female | No | Sun | Dinner | 4 |

In [29]:
```python
tips["total_bill"] = tips["total_bill"] - 2
tips["new_bill"] = tips["total_bill"] / 2
tips.head()
```

Out[29]:

| | total_bill | tip | sex | smoker | day | time | size | new_bill |
|---|---|---|---|---|---|---|---|---|
| 0 | 14.99 | 1.01 | Female | No | Sun | Dinner | 2 | 7.495 |
| 1 | 8.34 | 1.66 | Male | No | Sun | Dinner | 3 | 4.170 |
| 2 | 19.01 | 3.50 | Male | No | Sun | Dinner | 3 | 9.505 |
| 3 | 21.68 | 3.31 | Male | No | Sun | Dinner | 2 | 10.840 |
| 4 | 22.59 | 3.61 | Female | No | Sun | Dinner | 4 | 11.295 |

In [30]:
```python
tips = tips.drop("new_bill", axis=1)
```

In [31]:
```
tips[tips["total_bill"] > 10]
```

Out[31]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| **0** | 14.99 | 1.01 | Female | No | Sun | Dinner | 2 |
| **2** | 19.01 | 3.50 | Male | No | Sun | Dinner | 3 |
| **3** | 21.68 | 3.31 | Male | No | Sun | Dinner | 2 |
| **4** | 22.59 | 3.61 | Female | No | Sun | Dinner | 4 |
| **5** | 23.29 | 4.71 | Male | No | Sun | Dinner | 4 |
| **...** | ... | ... | ... | ... | ... | ... | ... |
| **239** | 27.03 | 5.92 | Male | No | Sat | Dinner | 3 |
| **240** | 25.18 | 2.00 | Female | Yes | Sat | Dinner | 2 |
| **241** | 20.67 | 2.00 | Male | Yes | Sat | Dinner | 2 |
| **242** | 15.82 | 1.75 | Male | No | Sat | Dinner | 2 |
| **243** | 16.78 | 3.00 | Female | No | Thur | Dinner | 2 |

204 rows × 7 columns

In [32]:
```
tips["bucket"] = np.where(tips["total_bill"] < 10, "low", "high")
tips
```

Out[32]:

| | total_bill | tip | sex | smoker | day | time | size | bucket |
|---|---|---|---|---|---|---|---|---|
| **0** | 14.99 | 1.01 | Female | No | Sun | Dinner | 2 | high |
| **1** | 8.34 | 1.66 | Male | No | Sun | Dinner | 3 | low |
| **2** | 19.01 | 3.50 | Male | No | Sun | Dinner | 3 | high |
| **3** | 21.68 | 3.31 | Male | No | Sun | Dinner | 2 | high |
| **4** | 22.59 | 3.61 | Female | No | Sun | Dinner | 4 | high |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... |
| **239** | 27.03 | 5.92 | Male | No | Sat | Dinner | 3 | high |
| **240** | 25.18 | 2.00 | Female | Yes | Sat | Dinner | 2 | high |
| **241** | 20.67 | 2.00 | Male | Yes | Sat | Dinner | 2 | high |
| **242** | 15.82 | 1.75 | Male | No | Sat | Dinner | 2 | high |
| **243** | 16.78 | 3.00 | Female | No | Thur | Dinner | 2 | high |

244 rows × 8 columns

In [33]:
```
tips["date1"] = pd.Timestamp("2013-01-15")
tips["date2"] = pd.Timestamp("2015-02-15")
tips["date1_year"] = tips["date1"].dt.year
tips["date2_month"] = tips["date2"].dt.month
tips["date1_next"] = tips["date1"] + pd.offsets.MonthBegin()
tips["months_between"] = tips["date2"].dt.to_period("M") - tips["date1"].dt.to_

tips[["date1", "date2", "date1_year", "date2_month", "date1_next", "months_betw
```

Out[33]:

| | date1 | date2 | date1_year | date2_month | date1_next | months_between |
|---|---|---|---|---|---|---|
| **0** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |
| **1** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |
| **2** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |
| **3** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |
| **4** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |
| **...** | ... | ... | ... | ... | ... | ... |
| **239** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |
| **240** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |
| **241** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |
| **242** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |
| **243** | 2013-01-15 | 2015-02-15 | 2013 | 2 | 2013-02-01 | <25 * MonthEnds> |

244 rows × 6 columns

In [34]:
```
tips[["sex", "total_bill", "tip"]]
```

Out[34]:

| | sex | total_bill | tip |
|---|---|---|---|
| **0** | Female | 14.99 | 1.01 |
| **1** | Male | 8.34 | 1.66 |
| **2** | Male | 19.01 | 3.50 |
| **3** | Male | 21.68 | 3.31 |
| **4** | Female | 22.59 | 3.61 |
| **...** | ... | ... | ... |
| **239** | Male | 27.03 | 5.92 |
| **240** | Female | 25.18 | 2.00 |
| **241** | Male | 20.67 | 2.00 |
| **242** | Male | 15.82 | 1.75 |
| **243** | Female | 16.78 | 3.00 |

244 rows × 3 columns

In [35]:
```
tips.drop("sex", axis=1)
```

Out[35]:

| | total_bill | tip | smoker | day | time | size | bucket | date1 | date2 | date1_year | date2_month |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 14.99 | 1.01 | No | Sun | Dinner | 2 | high | 2013-01-15 | 2015-02-15 | 2013 | 2 |
| **1** | 8.34 | 1.66 | No | Sun | Dinner | 3 | low | 2013-01-15 | 2015-02-15 | 2013 | 2 |
| **2** | 19.01 | 3.50 | No | Sun | Dinner | 3 | high | 2013-01-15 | 2015-02-15 | 2013 | 2 |
| **3** | 21.68 | 3.31 | No | Sun | Dinner | 2 | high | 2013-01-15 | 2015-02-15 | 2013 | 2 |
| **4** | 22.59 | 3.61 | No | Sun | Dinner | 4 | high | 2013-01-15 | 2015-02-15 | 2013 | 2 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **239** | 27.03 | 5.92 | No | Sat | Dinner | 3 | high | 2013-01-15 | 2015-02-15 | 2013 | 2 |
| **240** | 25.18 | 2.00 | Yes | Sat | Dinner | 2 | high | 2013-01-15 | 2015-02-15 | 2013 | 2 |
| **241** | 20.67 | 2.00 | Yes | Sat | Dinner | 2 | high | 2013-01-15 | 2015-02-15 | 2013 | 2 |
| **242** | 15.82 | 1.75 | No | Sat | Dinner | 2 | high | 2013-01-15 | 2015-02-15 | 2013 | 2 |
| **243** | 16.78 | 3.00 | No | Thur | Dinner | 2 | high | 2013-01-15 | 2015-02-15 | 2013 | 2 |

244 rows × 13 columns

In [40]:
```
tips.rename(columns={"total_bill": "total_bill_2"})
```

Out[40]:

| | total_bill_2 | tip | sex | smoker | day | time | size | bucket | date1 | date2 | date1_year | da |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **67** | 1.07 | 1.00 | Female | Yes | Sat | Dinner | 1 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **92** | 3.75 | 1.00 | Female | Yes | Fri | Dinner | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **111** | 5.25 | 1.00 | Female | No | Sat | Dinner | 1 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **145** | 6.35 | 1.50 | Female | No | Thur | Lunch | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **135** | 6.51 | 1.25 | Female | No | Thur | Lunch | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **182** | 43.35 | 3.50 | Male | Yes | Sun | Dinner | 3 | high | 2013-01-15 | 2015-02-15 | 2013 | |
| **156** | 46.17 | 5.00 | Male | No | Sun | Dinner | 6 | high | 2013-01-15 | 2015-02-15 | 2013 | |
| **59** | 46.27 | 6.73 | Male | No | Sat | Dinner | 4 | high | 2013-01-15 | 2015-02-15 | 2013 | |
| **212** | 46.33 | 9.00 | Male | No | Sat | Dinner | 4 | high | 2013-01-15 | 2015-02-15 | 2013 | |
| **170** | 48.81 | 10.00 | Male | Yes | Sat | Dinner | 3 | high | 2013-01-15 | 2015-02-15 | 2013 | |

244 rows × 14 columns

In [38]:
```
tips = tips.sort_values(["sex", "total_bill"])
tips.head(2)
```

Out[38]:

| | total_bill | tip | sex | smoker | day | time | size | bucket | date1 | date2 | date1_year | date2_m |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **67** | 1.07 | 1.0 | Female | Yes | Sat | Dinner | 1 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **92** | 3.75 | 1.0 | Female | Yes | Fri | Dinner | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |

In [41]:
```python
tips["time"].str.len()
```

Out[41]:
```
67     6
92     6
111    6
145    5
135    5
      ..
182    6
156    6
59     6
212    6
170    6
Name: time, Length: 244, dtype: int64
```

In [42]:
```python
tips["time"].str.rstrip().str.len()
```

Out[42]:
```
67     6
92     6
111    6
145    5
135    5
      ..
182    6
156    6
59     6
212    6
170    6
Name: time, Length: 244, dtype: int64
```

In [43]:
```python
tips["sex"].str.find("ale")
```

Out[43]:
```
67     3
92     3
111    3
145    3
135    3
      ..
182    1
156    1
59     1
212    1
170    1
Name: sex, Length: 244, dtype: int64
```

In [44]:
```
tips["sex"].str[0:1]
```

Out[44]:
```
67     F
92     F
111    F
145    F
135    F
      ..
182    M
156    M
59     M
212    M
170    M
Name: sex, Length: 244, dtype: object
```

In [45]:
```
pd.pivot_table(tips, values="tip", index=["size"], columns=["sex"], aggfunc=np.
```

Out[45]:

| sex | Female | Male |
|---|---|---|
| size | | |
| 1 | 1.276667 | 1.920000 |
| 2 | 2.528448 | 2.614184 |
| 3 | 3.250000 | 3.476667 |
| 4 | 4.021111 | 4.172143 |
| 5 | 5.140000 | 3.750000 |
| 6 | 4.600000 | 5.850000 |

In [54]:
```
tips.iloc[1:2,0:3]
```

Out[54]:

| | total_bill | tip | sex |
|---|---|---|---|
| 92 | 3.75 | 1.0 | Female |

In [56]:
```python
tips == "3.75"
```

Out[56]:

| | total_bill | tip | sex | smoker | day | time | size | bucket | date1 | date2 | date1_year | date2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **67** | False | False | False | False | False | False | False | False | False | False | False | |
| **92** | False | False | False | False | False | False | False | False | False | False | False | |
| **111** | False | False | False | False | False | False | False | False | False | False | False | |
| **145** | False | False | False | False | False | False | False | False | False | False | False | |
| **135** | False | False | False | False | False | False | False | False | False | False | False | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **182** | False | False | False | False | False | False | False | False | False | False | False | |
| **156** | False | False | False | False | False | False | False | False | False | False | False | |
| **59** | False | False | False | False | False | False | False | False | False | False | False | |
| **212** | False | False | False | False | False | False | False | False | False | False | False | |
| **170** | False | False | False | False | False | False | False | False | False | False | False | |

244 rows × 14 columns

In [57]:
```python
tips["day"].str.contains("S")
```

Out[57]:
```
67      True
92     False
111     True
145    False
135    False
       ...
182     True
156     True
59      True
212     True
170     True
Name: day, Length: 244, dtype: bool
```

In [58]:
```python
tips.replace("Thu", "Thursday")
```

Out[58]:

| | total_bill | tip | sex | smoker | day | time | size | bucket | date1 | date2 | date1_year | date |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **67** | 1.07 | 1.00 | Female | Yes | Sat | Dinner | 1 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **92** | 3.75 | 1.00 | Female | Yes | Fri | Dinner | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **111** | 5.25 | 1.00 | Female | No | Sat | Dinner | 1 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **145** | 6.35 | 1.50 | Female | No | Thur | Lunch | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **135** | 6.51 | 1.25 | Female | No | Thur | Lunch | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **182** | 43.35 | 3.50 | Male | Yes | Sun | Dinner | 3 | high | 2013-01-15 | 2015-02-15 | 2013 | |
| **156** | 46.17 | 5.00 | Male | No | Sun | Dinner | 6 | high | 2013-01-15 | 2015-02-15 | 2013 | |
| **59** | 46.27 | 6.73 | Male | No | Sat | Dinner | 4 | high | 2013-01-15 | 2015-02-15 | 2013 | |
| **212** | 46.33 | 9.00 | Male | No | Sat | Dinner | 4 | high | 2013-01-15 | 2015-02-15 | 2013 | |
| **170** | 48.81 | 10.00 | Male | Yes | Sat | Dinner | 3 | high | 2013-01-15 | 2015-02-15 | 2013 | |

244 rows × 14 columns

In [59]:
```python
tips_summed = tips.groupby(["sex", "smoker"])[["total_bill", "tip"]].sum()
tips_summed
```

Out[59]:

| | | total_bill | tip |
|---|---|---|---|
| **sex** | **smoker** | | |
| **Female** | **No** | 869.68 | 149.77 |
| | **Yes** | 527.27 | 96.74 |
| **Male** | **No** | 1725.75 | 302.00 |
| | **Yes** | 1217.07 | 183.07 |

In [61]:
```
gb = tips.groupby("smoker")["total_bill"]
tips["adj_total_bill"] = tips["total_bill"] - gb.transform("mean")
tips.head(2)
```

Out[61]:

| | total_bill | tip | sex | smoker | day | time | size | bucket | date1 | date2 | date1_year | date2_m |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **67** | 1.07 | 1.0 | Female | Yes | Sat | Dinner | 1 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **92** | 3.75 | 1.0 | Female | Yes | Fri | Dinner | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |

In [62]:
```
tips.groupby(["sex", "smoker"]).first()
```

Out[62]:

| | | total_bill | tip | day | time | size | bucket | date1 | date2 | date1_year | date2_mon |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **sex** | **smoker** | | | | | | | | | | |
| **Female** | **No** | 5.25 | 1.00 | Sat | Dinner | 1 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| | **Yes** | 1.07 | 1.00 | Sat | Dinner | 1 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| **Male** | **No** | 5.51 | 2.00 | Thur | Lunch | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |
| | **Yes** | 5.25 | 5.15 | Sun | Dinner | 2 | low | 2013-01-15 | 2015-02-15 | 2013 | |

In [64]:
```
url = ("https://raw.github.com/pandas-dev""/pandas/main/pandas/tests/io/data/cs

tips = pd.read_csv(url)
tips.head(2)
```

Out[64]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| **0** | 16.99 | 1.01 | Female | No | Sun | Dinner | 2 |
| **1** | 10.34 | 1.66 | Male | No | Sun | Dinner | 3 |

In [65]:
```
tips[tips["total_bill"] > 10]
```

Out[65]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| **0** | 16.99 | 1.01 | Female | No | Sun | Dinner | 2 |
| **1** | 10.34 | 1.66 | Male | No | Sun | Dinner | 3 |
| **2** | 21.01 | 3.50 | Male | No | Sun | Dinner | 3 |
| **3** | 23.68 | 3.31 | Male | No | Sun | Dinner | 2 |
| **4** | 24.59 | 3.61 | Female | No | Sun | Dinner | 4 |
| **...** | ... | ... | ... | ... | ... | ... | ... |
| **239** | 29.03 | 5.92 | Male | No | Sat | Dinner | 3 |
| **240** | 27.18 | 2.00 | Female | Yes | Sat | Dinner | 2 |
| **241** | 22.67 | 2.00 | Male | Yes | Sat | Dinner | 2 |
| **242** | 17.82 | 1.75 | Male | No | Sat | Dinner | 2 |
| **243** | 18.78 | 3.00 | Female | No | Thur | Dinner | 2 |

227 rows × 7 columns

In [66]:
```
tips[["sex", "total_bill", "tip"]]
```

Out[66]:

| | sex | total_bill | tip |
|---|---|---|---|
| **0** | Female | 16.99 | 1.01 |
| **1** | Male | 10.34 | 1.66 |
| **2** | Male | 21.01 | 3.50 |
| **3** | Male | 23.68 | 3.31 |
| **4** | Female | 24.59 | 3.61 |
| **...** | ... | ... | ... |
| **239** | Male | 29.03 | 5.92 |
| **240** | Female | 27.18 | 2.00 |
| **241** | Male | 22.67 | 2.00 |
| **242** | Male | 17.82 | 1.75 |
| **243** | Female | 18.78 | 3.00 |

244 rows × 3 columns

In [67]:
```python
tips = tips.sort_values(["sex", "total_bill"])
tips
```

Out[67]:

| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| **67** | 3.07 | 1.00 | Female | Yes | Sat | Dinner | 1 |
| **92** | 5.75 | 1.00 | Female | Yes | Fri | Dinner | 2 |
| **111** | 7.25 | 1.00 | Female | No | Sat | Dinner | 1 |
| **145** | 8.35 | 1.50 | Female | No | Thur | Lunch | 2 |
| **135** | 8.51 | 1.25 | Female | No | Thur | Lunch | 2 |
| **...** | ... | ... | ... | ... | ... | ... | ... |
| **182** | 45.35 | 3.50 | Male | Yes | Sun | Dinner | 3 |
| **156** | 48.17 | 5.00 | Male | No | Sun | Dinner | 6 |
| **59** | 48.27 | 6.73 | Male | No | Sat | Dinner | 4 |
| **212** | 48.33 | 9.00 | Male | No | Sat | Dinner | 4 |
| **170** | 50.81 | 10.00 | Male | Yes | Sat | Dinner | 3 |

244 rows × 7 columns

In [5]:
```python
print(tips.iloc[-20:, :12].to_string())
```

```
     total_bill  tip     sex smoker  day    time  size
224       13.42  1.58    Male    Yes  Fri   Lunch     2
225       16.27  2.50  Female    Yes  Fri   Lunch     2
226       10.09  2.00  Female    Yes  Fri   Lunch     2
227       20.45  3.00    Male     No  Sat  Dinner     4
228       13.28  2.72    Male     No  Sat  Dinner     2
229       22.12  2.88  Female    Yes  Sat  Dinner     2
230       24.01  2.00    Male    Yes  Sat  Dinner     4
231       15.69  3.00    Male    Yes  Sat  Dinner     3
232       11.61  3.39    Male     No  Sat  Dinner     2
233       10.77  1.47    Male     No  Sat  Dinner     2
234       15.53  3.00    Male    Yes  Sat  Dinner     2
235       10.07  1.25    Male     No  Sat  Dinner     2
236       12.60  1.00    Male    Yes  Sat  Dinner     2
237       32.83  1.17    Male    Yes  Sat  Dinner     2
238       35.83  4.67  Female     No  Sat  Dinner     3
239       29.03  5.92    Male     No  Sat  Dinner     3
240       27.18  2.00  Female    Yes  Sat  Dinner     2
241       22.67  2.00    Male    Yes  Sat  Dinner     2
242       17.82  1.75    Male     No  Sat  Dinner     2
243       18.78  3.00  Female     No Thur  Dinner     2
```

**Syed Afroz Ali**

In [ ]: